Neural Moderation of ASMR Erotica Content in Social Networks

Yixin Chen, Di Jiang, Conghui Tan, Yuanfeng Song, *Member, IEEE*, Chen Zhang, *Member, IEEE*, and Lei Chen, *Fellow, IEEE*

***Abstract*—With the popularity of video/audio streaming applications in recent years, the wide spread of Autonomous Sensory Meridian Response (ASMR) erotica content is becoming a serious issue in social networks. Due to the subtle nature of ASMR erotica and its relative rareness in real scenario, detecting ASMR erotica contents is a challenging task. In this article, we propose a novel neural framework for ASMR erotica content moderation. The proposed framework consists of a pipeline of novel strategies to tackle challenges unique in ASMR Erotica Contents such as data scarcity and imbalanced data. Based on large-scale industrial data, the proposed framework demonstrates high moderation accuracy in quantitative analysis and significantly outperforming the existing counterparts.**

***Index Terms*—ASMR, content moderation, neural network.**

I. Introduction

We are in an era of citizen media. Countless user-generated content (UGC) is created and shared on the internet via social media such as Facebook, Instagram, and Youtube. However, a small fraction of these user-generated content is illegal, such as offensive language, hate speeches and pornographic videos. For social networks, these illegal content severely damage the reputation of the platform or even bring legal punishment from the authority. Therefore, systematic moderation strategies are necessary for contemporary social networks. Among the wide varieties of illegal contents, Autonomous Sensory Meridian Response (ASMR) erotica audio is the hardest to detect automatically, since it is not necessarily associated with human languages and cannot be effectively processed by existing automatic speech recognition (ASR) or natural language processing (NLP) techniques. Therefore, we proposed a novel end-to-end neural architecture for ASMR erotica moderation.

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In this article, we work towards more effective methods to detect ASMR erotica audios. There are some unique challenges in this task. First, ASMR erotica audios only take a small fraction of the UGC in social networks, so it is hard to collect sufficient training samples, which further results in class-unbalanced problems. Second, due to the repetitive nature of the ASMR erotic signals, we employ Convolutional neural network (CNN) in this work since it is superior in capturing the local pattern in the repetitive signals [1]. However, how to measure suitable architecture hyper-parameters of the CNN with a low computing resource is still an open question.

To solve the aforementioned problems, we design a novel neural framework that seamlessly integrates various advanced techniques, including data augmentation, neural architecture search (NAS), and class reweighting, towards an industry-level solution to this ASMR erotica audios detection problem. In this framework, data augmentation is used to solve the data scarcity problem, while NAS can automatically find better neural network architecture with low computing resources. Finally, class reweighting will empower the model to make full use of the classes with fewer samples when the data is unbalanced. Experiments on real-life datasets verify the superiority of our solution over existing methods.

The rest of this article is organized as follows. Section II reviews the most relevant literature. Section III describes the neural architecture search process and the model architecture. Section IV reports the experimental results. Finally, Section V concludes this paper.

II. Related Work

A variety of signal processing and machine learning techniques have been applied to audio classification, including matrix factorization [2], [3], dictionary learning [4], [5], wavelet filter banks [6] and deep convolutional neural networks. Convolutional neural network (CNN) is good at extracting local patterns, thus it has been widely applied in the audio field, such as acoustic event detection [7], music start detection [8], automatic speech recognition [9], speech-based emotion recognition [10], snore sound recognition [11] and autism severity classification [12].

For class imbalance problems in deep learning, the mainstream methods can be categorized into data-level techniques, algorithm-level methods, and hybrid approaches. Data-level techniques balance distributions by generating new objects for

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minority groups (oversampling) [13] and removing examples for majority groups (undersampling) [14]. Algorithm-level methods [15] aim to alleviate the bias towards majority objects by changing the model. Hybrid approaches [16] combine the data-level methods and algorithm-level methods, which result in a robust and efficient model.

Designing a high-performance neural network requires much professional knowledge, many experiments, and high cost, which limits the application of neural networks. Under such circumstances, Neural architecture search (NAS) was proposed to automatically design a high-performance network architecture according to sample sets. It can even find some structures that have not been proposed before. Based on different principles, NAS can be divided into methods based on reinforcement learning [17], [18], methods based on gradient [19], [20] and methods based on evolutionary algorithm [21], [22].

Our work is different from the aforementioned ones in the following ways. First, labeled ASMR data is typically difficult to obtain. Data augmentation can generate additional training data by using some deformations while maintaining semantic validity labels. By training the network on the additional augmented data, more data can be obtained and the invisible data can be better generalized. Data augmentation has been proven to improve the model accuracy of music classification tasks [23]. Second, finding suitable architecture and hyper-parameters of CNN requires much experience and practice, NAS can automatically design a network structure with good performance by limited computing resources. NAS is successfully used in acoustic scene classification in [24]. Compared with other optimization methods, NAS based on our proposed evolutionary algorithm is much more efficient and flexible since the mutation and crossover operations generate random candidates and the fitness function can be designed arbitrarily, which can further generate more robust and precise model. Lastly, rareness of ASMR erotica will heavily affect the accuracy of model, changing class weight [25] and ratio of training data in the training process can make model pay more attention to the class with small samples. Compared with data-level techniques and algorithm-level methods, hybrid approaches we used combine the advantages and reduce the weakness, resulting in robust and efficient model.

III. Neural Moderation

A. Data Processing

Voice is generated by the physiological movement of the voice organ under the control of the brain. With some digital signal processing algorithms, voice features can be more accurately obtained, thereby contributing to subsequent voice tasks. Fig. 1(a) shows an example of an ASMR erotica signal. In the time dimension, the ASMR erotica signal is composed of regular repetitive arrangement of signal primitives. For ASMR erotica moderation, we slice the audio signal to overlapping frames of window length of 64 ms (1024 samples for 16 kHz audio sample rate) and hop size of 16 ms. For each frame, we extract the Mel Frequency Cepstrum Coefficient (MFCC), which will work as the input features of the classification model.

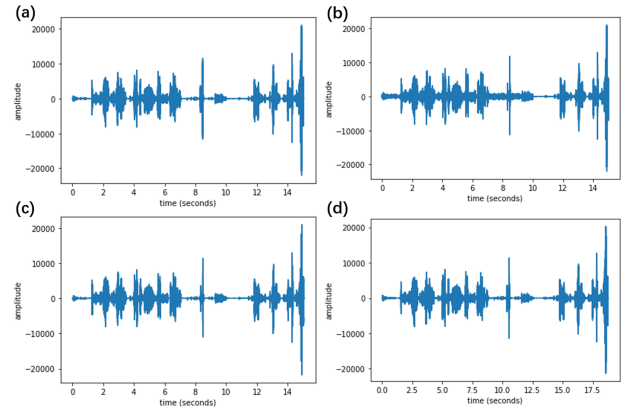

Fig. 1. (a) A waveform of a ASMR signal; (b) the waveform after noise augmentation; (c) the waveform after changing pitch; (d) the waveform after changing speed.

Considering that the number of ASMR erotica samples are usually small, we adopt data augmentation to generate more illegal audios before feature extraction. Here we use three different strategies:

- *Noise augmentation:* mix audio with another audio containing background sounds from different types of acoustic scenes. We use three types of noise: white noise, music, and audios extracted from online videos. For each illegal sample, we randomly draw multiple noises, and mix each noise with the original sample to generate a new sample. The mixed audio is generated by $mix_audio = audio + w \cdot noise$ with random weight $w \in [0.1, 0.3]$.
- *Pitch change:* increase or decrease the pitch while keep the speed unchanged. Each sample is pitch shifted by 4 different values (in semitones): -2, -1, 1, 2.
- *Speed change:* speed up or slow down the audio while keep the pitch unchanged. The speed for each sample is perturbed by distinct factors: 0.8, 0.9, 1.1, 1.2.

In Fig. 1(a)–(d), we give an example of ASMR waveform, waveform of noise augmented, waveform of changing pitch and waveform of changing speed, respectively.

B. Moderation Architecture Search

Due to the repetitive nature of ASMR erotic signal, we choose convolutional neural network (CNN) as the primary architecture. Compared to RNN-based models like LSTM, convolution over the time domain does better in capturing the local pattern. For CNN, numerous advanced variants have been proposed in recent years, such as residual network and efficient Net, which show promising results on many tasks. However, these variants have deep network structure and large number of parameters, making them unsuitable for our tasks whose data size is limited. Hence, we explore better CNN structure for this task via neural architecture search (NAS).

The goal of NAS for CNN is seeking for better architecture hyper-parameters (number of convolutional layers, number of filters, etc.) and better learning hyper-parameters (learning rate, batch size, dropout probability, regularization coefficient, etc.). The number of potential combinations of these hyper-parameters

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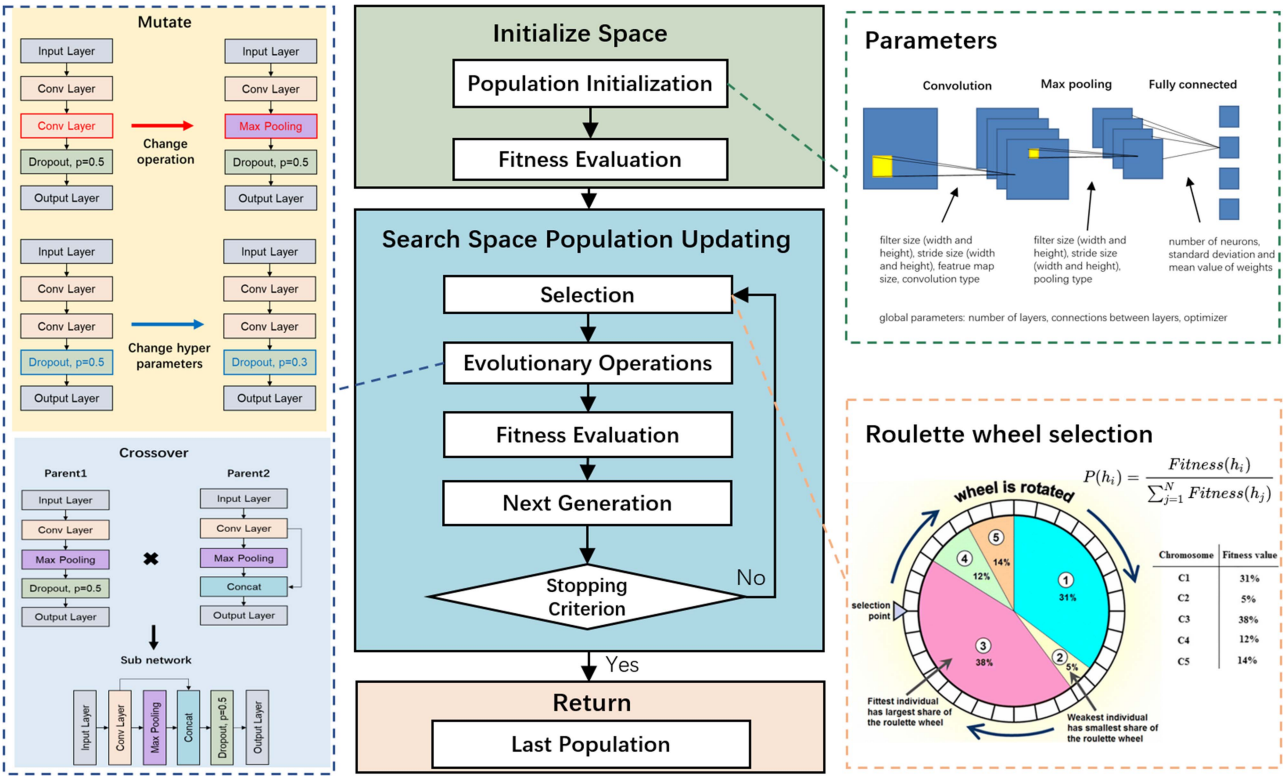

Fig. 2. The process of neural architecture search, consisting of population initialization, selection and evolutionary.

are too large for manual selection. Hence, we use neural architecture search based on evolutionary algorithm to find the best combination of network hyper-parameters.

The searching process is shown in Fig. 2. We first randomly initialize a number of network configurations as initial candidates. In each iteration, we train each network on the training set, and compute its prediction accuracy on the validation set, which acts as the fitness for this network architecture. Then, we select some networks by roulette wheel selection, where the selection probability of a network is proportional to its fitness. Finally, we generate new network candidates by randomly mutating and crossovering the selected networks. We repeat this evaluation-selection-generation process for a fixed number of iterations.

For neural architecture search, because the two dimensions of acoustic features are quite different, we choose convolution kernels with different sizes in different dimensions and different pooling sizes in different dimensions. The convolution kernel size is set to 3×5 and the pooling size is set to 2×4. For the other hyper-parameters, we decide their values via NAS and limit their ranges to reduce the complexity of NAS: we set the maximum convolutional layer as 3, and the number of filters is between is 8 and 96, the maximum fully connected layer is 2, and the dimension of the fully connected layer is between 8 and 96.

For NAS, we run it with 10 iterations. Its initial population size is set to 10. When evaluating the fitness, each model is trained for 3 epochs and the checkpoint with the best validation accuracy is chosen. The roulette wheel selection will choose 20% of the current models.

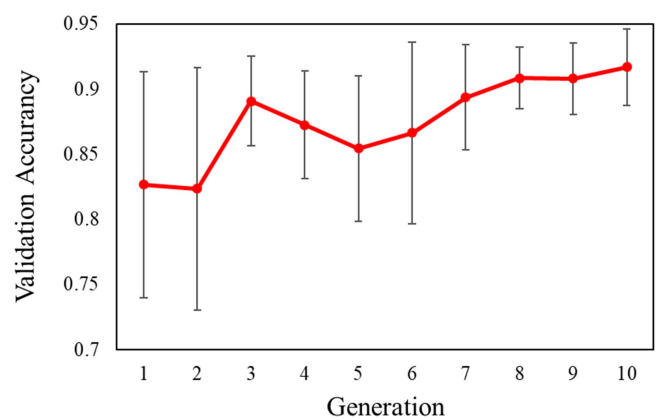

Fig. 3. Validation accuracy of every generation.

The validation accuracy of the best network in each generation is shown in Fig. 3. As the algorithm iterates, the population becomes better and more stable. The best model configuration is obtained at the final generation. It is confirmed that we can find better network parameters by NAS.

The best convolutional neural network we obtain from NAS is composed of 3 convolutional layers interleaved with 2 pooling operations, followed by 2 fully connected layers. The overall architecture is presented in Fig. 4. More details about this architecture are as follows:

- The first layer includes convolution with 12 filters and 3×5 kernel size for the time and frequency dimensions, respectively, followed by a 2×4 strided max-pooling.

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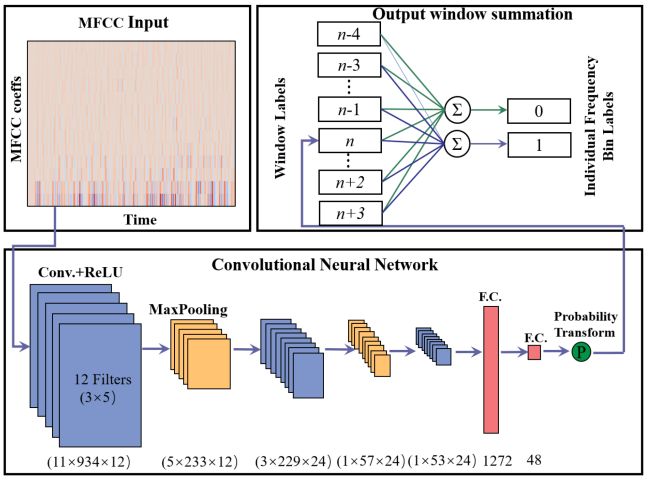

Fig. 4. Neural Architecture is composed of 3 convolution layers, 3 pooling layers, a fully connected layer and an output layer.

ℓ_2-regularization with a penalty coefficient of 0.0005 and a rectified linear unit (ReLU) activation function is applied in convolutional layer.

The second layer is similar to the first layer, but the number of filters is increased to 24.

- The third layer is also similar. The differences are that, the number of filters becomes 48, and no pooling is applied.
- The fourth layer is a fully connected layer, whose output dimension is 48 and activation function is ReLU. ℓ_2 regularization is also applied with a factor of 0.001.
- The output layer is also a fully connected layer with 2 output units, softmax activation function and 0.001 ℓ_2 regularization.

Since it is a binary classification problem, it is natural to use cross entropy as the loss function to train the model.

C. Class Unbalance

In practice, ASMR erotica audios are rare. In our real scenario, such audios take less than 1% of all the audios. So, this classification problem is heavily unbalanced, which brings difficulties for both training and inference.

To deal with class unbalance, we study three different strategies:

- Changing the training data ratio by adding more normal audios into the training data;
- Assigning larger weight for the erotica audios in the training loss function;
- Altering classification threshold in the inference phase.

Actually, different strategies can be used together to achieve better performance. For example, one can use distinct class weights in the training phase, and also perturb the classification threshold for inference. But trying all possible strategy combinations is too costly. Our methodology is to use equal error rate (ERR) as the evaluation metric. ERR is the false positive rate and false negative rate when the classification threshold is chosen so that these two rates equal. The use of ERR eliminates the impact of classification threshold. We can first compare the first two strategies with ERR, and then tune the classification threshold to maximize the target metric.

Fig. 5. The error rate of different class weights/different ratio training set.

IV. Experiments

A. Experimental Setup

In this section, we conduct experiments to verify the effectiveness of our proposed method.

All data used in our experiments are collected in a real industry scenario. The training data are composed of 33896 illegal ASMR erotica audios, and large number of normal audios since the normal audios are easier to gather. But in the experiment, we limit the number of normal audios to 200654, and make use of all these normal audios by default.

To ensure the test set can reflect the model performance in a real application, we directly sampled test data from an online business platform and then tag labels manually to simulate the distribution of real data. The test set we gathered includes 4122 illegal audios and 652801 normal audios, with a ratio of roughly proportionality to 1:158. The sampling rate of all these audios is 16 kHz and the maximum duration is 15 s. For audios shorter than 15 s, we padded zeros to make the duration consistent.

We will use precision, recall and F1 as the evaluation metric. By regarding ASMR erotica audios as the positive class, precision and recall are defined in [26].

B. Study on Class Unbalance

In this section, we present experiment results concerning the schemes for handling class unbalance. As we stated in Section III-C, we will first use ERR as the performance metric and test the other two strategies. The model we test here is the best model found by NAS.

We first vary the class weights of illegal and normal data in the training loss. In this experiment, we only use the same number of normal audios as the illegal ones. Since most of the audios are normal in practice, we assign larger weights for normal audios. The results are shown in Fig. 5. It can be observed that slightly increasing the class weight to 1:2 and 1:3 is helpful for improving performance, and weight 1:2 achieves the best performance. But when the weights for the normal data becomes too large, the ERRs deteriorates.

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TABLE I
THE RECALL, PRECISION AND F1 OF DIFFERENT CLASSIFICATION THRESHOLDS

threshold	recall	precision	F1
0.5	11.45%	94.73%	20.43%
0.8	22.48%	90.75%	36.03%
0.9	35.40%	87.77%	50.45%
0.99	83.91%	75.95%	79.74%
0.999	100.00%	59.43%	74.55%

TABLE II
THE RECALL, PRECISION AND F1 WITHOUT/WITH DATA AUGMENTATION

data augmentation	precision	recall	F1
Without data augmentation	85.44%	71.17%	77.65%
With data augmentation	86.22%	75.93%	80.75%

TABLE III
THE ERROR RATE AND PRECISION, RECALL, F1, SPEED WHEN THRESHOLD IS 0.99 OF DIFFERENT NETWORK STRUCTURE

model	ERR	precision	recall	F1	speed
Resnet 18	40.32%	11.30%	83.01%	19.89%	0.39s
Resnet 34	38.03%	12.40%	83.45%	21.59%	1.41s
[1]	29.98%	13.66%	87.89%	23.64%	0.15s
Our model	20.36%	86.22%	75.93%	80.75%	0.15s

Moreover, we fix the class weights and vary the ratio between the illegal and normal data in the training set, by adding more normal data into it. The results are in Fig. 5. Clearly, adding more normal data is beneficial for improving performance, and its effect is much stronger than changing class weights (ERR 20.36% versus 33.39%). Hence, we can conclude that adding more normal data is a better strategy than changing class weights.

By fixing the data ratio to 1:6, we now swap to the problem of determining classification threshold. We vary the classification thresholds and report the corresponding metrics in Table I. With a larger threshold, the recall is improving, while the precision is decreasing. The F1 value, which integrates both recall and precision, attains the best value with threshold $= 0.99$. Therefore, we adopt threshold $= 0.99$ by default in the following experiments.

C. Effectiveness of Data Augmentation and NAS

In this part, we conduct experiments to verify the effectiveness of data augmentation and neural architecture search.

In Table II, we report the metrics for the models trained with data augmentation and without augmentation. The model structure adopted for both settings is the best architecture found by NAS. It is surprising that data augmentation on the illegal data is beneficial for both recall and precision. This suggests that adding more illegal training samples, even though they are faked, can boost the classification capacity.

In Table III, we compare our best model configuration from NAS with state-of-the-art model structures [1], [27] for audio classification. Data augmentation is applied for both models. Results show that our model is superior than the existing approach by relatively lower error rate and faster calculation speed, further proving the effectiveness of our NAS scheme.

V. CONCLUSION

In this article, we proposed a novel neural architecture for ASMR erotica moderation. Due to the rareness of ASMR erotica, training samples are hard to obtain and the problem is heavily class unbalanced. Data augmentation, neural architecture search and class reweighting are proposed to solve the problem. Experiments on real data set prove the superiority of our model. We wish that the framework can help the technique of speech content moderation to be utilized in a more proper and convenient manner in industrial scenarios.

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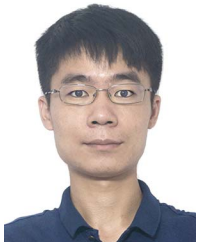

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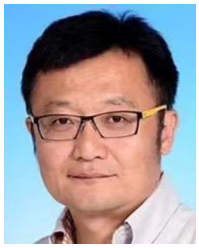

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